

Final assignment for the course Data visualization

Master Life Science Informatics

Type: Portfolio

The final mark is the result of two different and mandatory components:

- An original study based on data science methods (70%)
- A presentation (30%)

The original case study should be first presented in the seminar in the lecture on the 22nd of April. The first presentation is not graded. The format is a 3-minute presentation that should contain:

- Description of datasets
- Research Question & Hypothesis
- Expected use of Data visualization methods

The final study and the presentation can be completed individually or in groups of maximum three students. The final presentation will take place on Saturday 18th of July. The format is a 15-minute presentation followed by a 15-minute Q&A session.

Topic:

The topic is of free-election and must be presented during the specific lecture sessions of the course.

Expected Methods:

- Methods from descriptive statistics: Descriptive Statistics
- Statistics for scientific work: Literate Statistical Analysis
- Understanding of principles of data visualization & techniques
- Advanced Data visualization techniques: ggplot2

Introduction:

You act as a data analyst consultant for a firm working on the field of life sciences, for example: a company in the wearables sector that wants to enter the IoT market. As an expert on Life Science Informatics, you are required to analyse a database relevant for your company core business with the aim of providing clients with an overview on the R&D the company is doing.

An example could be how Apple used the mobility trends during the Covid-19 pandemic from IoT devices. The task would be to demonstrate how valuable the data is to clients in the field of coaching in a particular country/region.

Example of a database: Apple's mobility trend reports show how human mobility has changed in countries and cities worldwide since January 2020 and are based on location data of Apple's "maps" services. It is designed to help mitigate the spread of COVID-19, provide governments, research institutions, health authorities, and the general public with insights on the effects on human mobility of national and regional lockdown policies. The data covers 63 countries, 530

provinces and all major cities worldwide. (Source: <https://www.apple.com/covid19/mobility>). Additional potential databases to use with the Apple dataset: Census data/regional statistics/market trends/policy mechanisms...

Role played: You have a very good overview of the data you have at the company. You want to convince your team leader that the R&D department in which you work can win new potential clients by creating a new line of services/products. You can actually lead a small team and create a new department and a new line of services. You already had look into the data and found some very interesting patterns, that is the base of the service/product you want to produce.

Your role is to create a small report to present to your supervisor. Your data report should inform new clients about the scientific background behind the development of that new service/product or adapt existing ones. The question for you is therefore how to convince your audience that your idea is worth the time.

You know that your team leader is very busy and do not listen to new ideas at least they are well-developed and can bring real value to the company. You want to make sure that he/she allocates the time and resources you need to further develop the idea. Also, you know that the decision is not only made by your team leader, and your team leader will need to get approval from the R&D department chief officials. Therefore, it is important that you efficiently transmit why they should pay attention/invest in your idea.

Based on this description you must prepare a presentation (power point slides or any other format) to convince your supervisor, for which you have 15 minutes and one R/Python notebook with scientifically oriented analysis (3000 words, without tables or literature).

Style and grammar guidelines: We will be using the 7th APA guidelines, they are available online (<https://apastyle.apa.org/style-grammar-guidelines>) and cover paper format, in-text citations, Style, Bias free language, tables and figures and references.

Criteria:

- Good research question and result
- Good contextualisation of the statistical analysis in the selected region
- Quality and diversity of the graphics used
- Quality of the presentation: engaging capacity and clarity of the message.
- Using statistical key figures correctly
- Use and explanation of (advanced) statistical and visualization techniques not covered in the course.

Formal requirements for the final assignment project

Title page

Logo, subject, subject, mat no.

Overview

Short summary, keywords

List of contents, figures, tables, listings, abbreviations ...

1. Introduction

Short description of the topic and transition to the Problem/State of the Art.

2. Problem definition

State-of-the-Art

Introduction to the problem

3. Objectives

Brief formulation of the objectives of this work.

4. Methods

Description of the method and different Steps, what has been done? Why it is done in that way?

5. Results

Presentation and interpretation of the results.

6. Discussion

Pro- and contra discussion on results and the work.

7. Conclusions

Concluding words and a brief outlook on what else could be done or what the results could be used for.

Literature

Citations, online sources, data, images, ...

Appendix

Maps, source code, files...